

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2016-2017

MATHEMATICS FOR COMPUTER SCIENTISTS

Time allowed ONE Hour

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all the THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Questions on Logic, Sets, Functions, and Relations.

(a) Let $A = \{1,2,3,4\}$ and $B = \{2,4,5\}$ be two sets, and $P(A)$ and $P(B)$ denote their power sets respectively.

- (i) Calculate the set difference $A - B$, the set union $A \cup B$, the set intersection $A \cap B$, and the power set $P(B)$, respectively.

[4 marks]

- (ii) Let f be a function from A to B such that $f(1) = 2, f(2) = 4, f(3) = 2, f(4) = 5$. Let g be a function from B to A such that $g(2) = 2, g(4) = 1, g(5) = 4$. For each of these functions determine whether it is one-to-one, onto. Determine the function compositions $f \circ g$ and $g \circ f$ respectively. Show your work.

[6 marks]

- (iii) Let $R = \{(a, b) | a \in A \wedge b \in A \wedge a \leq (b - 1)\}$ be a subset of the Cartesian product $A \times A$. List all the elements in R .

[6 marks]

- (iv) Let $R_1 = \{(X, Y) | X \in P(A) \wedge Y \in P(B) \wedge X \in Y\}$ and $R_2 = \{(X, Y) | X \in P(A) \wedge Y \in P(B) \wedge X \subseteq Y\}$ be two subsets of the Cartesian product $P(A) \times P(B)$. Is R_1 empty? Is R_2 empty? Explain your answers.

[4 marks]

(b) Let A be the set of all people on Facebook. Let R be a relation on the set A such that $\forall a, b \in A, (a, b) \in R$ if and only if a shares at least one common friend with b on Facebook.

- (i) Is R reflexive, symmetric, and/or transitive? Explain your answers.

[6 marks]

- (ii) When is an ordered pair (a, b) in R^2 ? Explain your answer.

[2 marks]

- (iii) Let S be a relation on the set A such that $\forall a, b \in A, (a, b) \in S$ if and only if a knows about b . Determine $R \cup S, R \cap S$, and $R - S$.

[6 marks]

2. Questions on Logic, Number Theory, Induction, and Matrices.

(a) If a and b are integers with $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$. The notation $a \mid b$ denotes that a divides b . If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $a - b$. We use the notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m .

(i) Using $\forall, \exists, \neg, \wedge, \vee, \rightarrow$, and/or $\leftrightarrow$, express the above definition " a divides b " in predicate logic.
[4 marks]

(ii) Show that if a, b, c and m are integers such that $m \geq 2$, $c > 0$, and $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{mc}$.
[5 marks]

(iii) Let $a = bq + r$, where a, b, q and r are integers. The Euclidean algorithm says that the greatest common divisor of a and b is the same as that of b and r , denoted as $\gcd(a, b) = \gcd(b, r)$. Find the greatest common divisor of 91 and 287 using the Euclidean algorithm.
[5 marks]

(b) A sequence is an ordered list. For example, 3, 5, 8, 12, 17, 23, 30, 38, 47, ... is an infinite sequence.

(i) Provide a simple formula that generates an integer sequence that begins with the above given list.
[4 marks]

(ii) Let $\{a_n\}$ denote the above given sequence, where $n \geq 1$ are integers and $a_1 = 3$. Let $S_m = \sum_{j=m}^{2m-1} a_j$, where $m \geq 1$ are integers. List the first **THREE** terms S_1 , S_2 , and S_3 of the sequence $\{S_m\}$. Show your work.
[4 marks]

(iii) Suppose that A and B are square matrices with the property $AB = BA$. Using the mathematical induction show that $AB^n = B^nA$ for every positive integer n .
[6 marks]

(iv) Let $P(n)$ be the statement that $[(p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \dots \wedge (p_{n-1} \rightarrow p_n)] \rightarrow [(p_1 \wedge p_2 \wedge \dots \wedge p_{n-1}) \rightarrow p_n]$ is a tautology, where n is an integer. We can prove by mathematical induction that $P(n)$ holds whenever $p_1, p_2, \dots, p_n$ are propositions, where $n \geq 2$. Specify the statement that is used in the basic step of the mathematical induction and prove that it is true.
[5 marks]

3. Questions on Sets, Functions, Counting, and Probability.

(a) Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d, e\}$ be two sets, and a function $f: A \rightarrow B$ is defined such that every element in A is mapped to an element in B .

(i) How many such functions are there from A to B ? Show your work.

[4 marks]

(ii) Among the results in (i), how many functions are one-to-one? Show your work.

[5 marks]

(iii) Among the results in (i), how many functions are there that assign a to exactly one of the positive integers less than 3? Show your work.

[6 marks]

(iv) Assume that A is given such that $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. B and f remain the same as specified above. Using the Pigeonhole Principle show that f is not a one-to-one function.

[6 marks]

(b) What is the probability of the following events when we randomly select a permutation of the 5 lowercase letters of the English alphabet $\{a, b, c, d, e\}$?

(i) The first 3 letters of the permutation are in alphabetical order.

[6 marks]

(ii) a and e are next to each other in the permutation.

[6 marks]